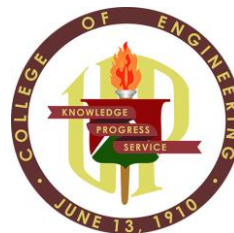
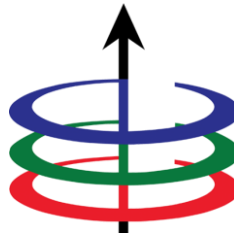


University of the Philippines Diliman
College of Engineering
Electrical and Electronics Engineering Institute

Linear Waveshaping

1st Semester, A.Y. 2026–2027



Student Outcomes

- Formulate mathematical expressions and sketch the output waveforms of first-order RC networks subjected to step, pulse, and rectangular inputs.
- Calculate critical parameters such as cutoff frequencies and tilt.
- Explain the conditions under which RC circuits function as linear waveshaping operators, particularly as differentiators and integrators.

Lecture Outline

- High Pass RC Circuit
- Low Pass RC Circuit

Linear Waveshaping

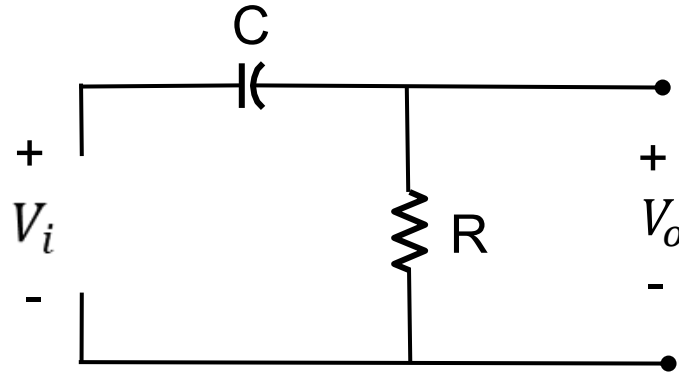
- Process whereby the form of a non-sinusoidal signal is altered by transmission through a linear network.
- Response of step, pulse square, ramp and exponential waveforms to certain simple RC, RL and RLC circuits
- Examples of linear waveshaping circuits we will be looking at:
 - High Pass RC
 - Low Pass RC

High Pass RC Circuit

Sinusoidal Input

$$V_i = A \cos(2\pi f t + \varphi)$$

$$X_c = \frac{1}{\omega C}$$



Parameters

- lower 3dB frequency is $f_L = \frac{1}{2\pi RC}$

- gain is $|H(f)| = \frac{1}{\sqrt{1 + \left(\frac{f_L}{f}\right)^2}}$



- the output leads the input by an angle of $\theta = \tan^{-1} \left(\frac{f_L}{f} \right)$

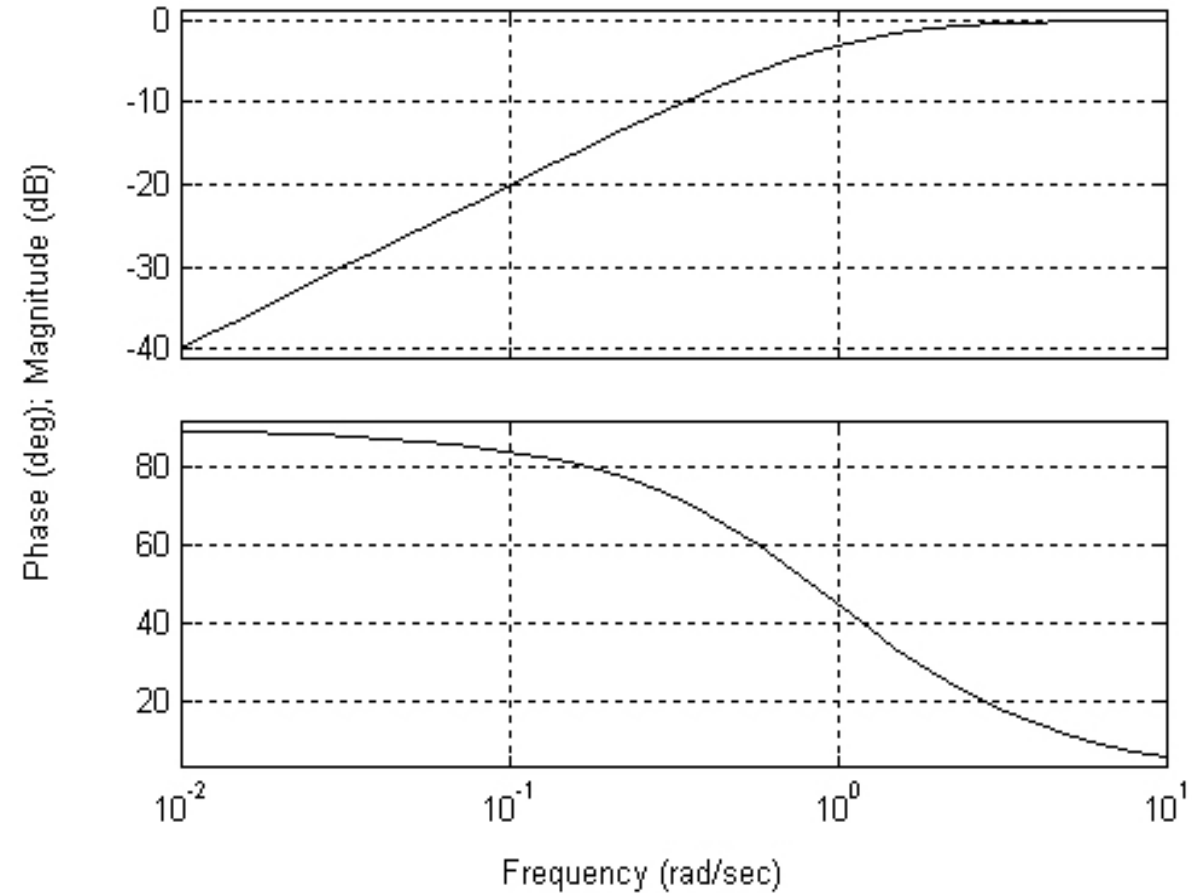
High Pass RC Circuit

Sinusoidal Input

Frequency Response

At $f = f_c$,

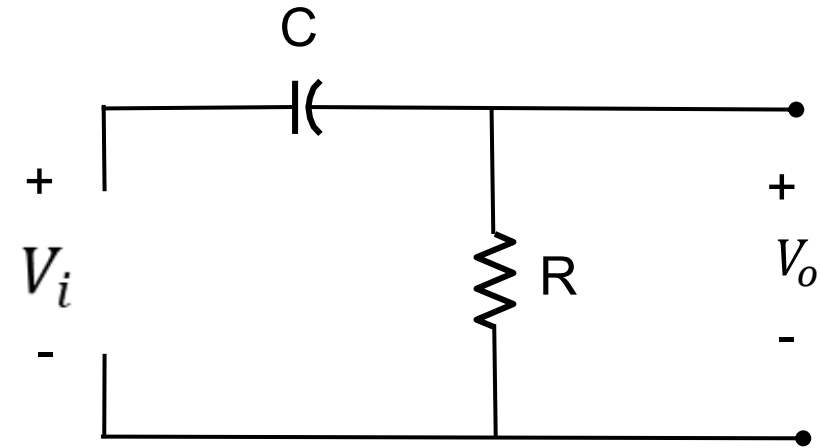
- $|H(f_c)| = 0.707$
- $X_c = R$



High Pass RC Circuit

Step Voltage Input

$$V_o = V_o(\infty) + [V_o(0^+) - V_o(\infty)]e^{-\frac{t}{RC}}$$

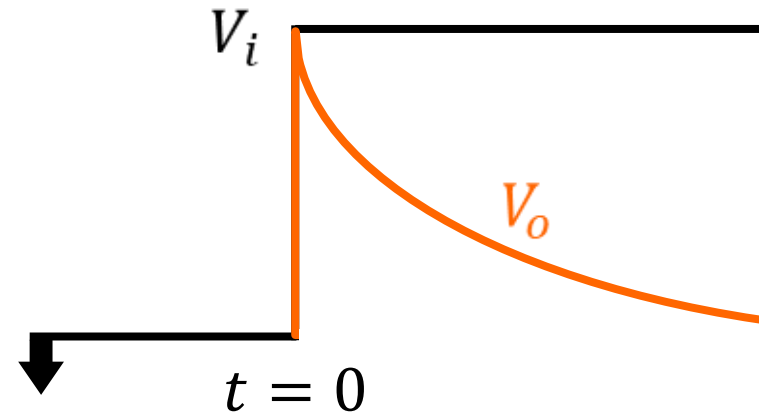


Steady-state analysis will show that

- $V_o(0^+) = V_i$
- $V_o(\infty) = 0$

Hence,

$$V_o = V_i e^{-\frac{t}{RC}}$$



t/RC	V_o/V_i
0.5	0.607
1	0.368
2	0.135
3	0.050
4	0.018
5	0.007

High Pass RC Circuit

Pulse Input

At $0 < t < t_p$

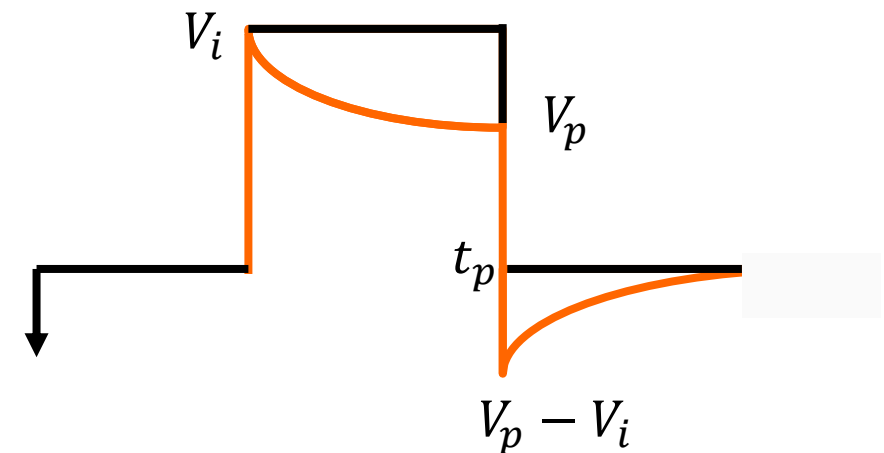
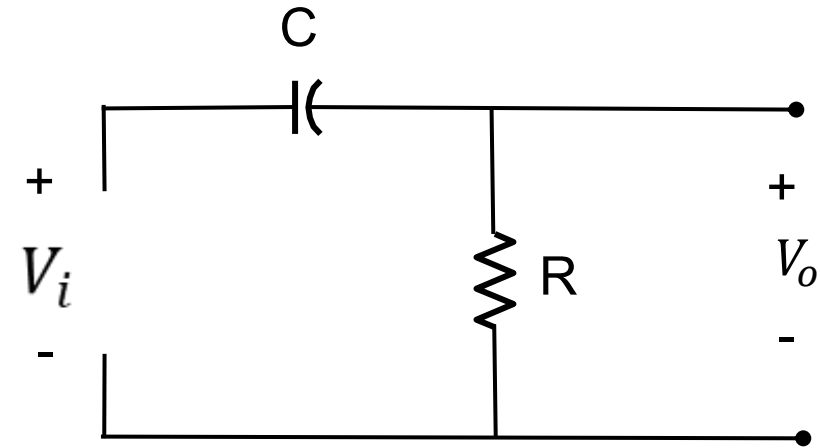
$$V_o = V_i e^{-\frac{t}{RC}}$$

At $t > t_p$

$$\begin{aligned} V_o &= \left(V_i e^{-\frac{t_p}{RC}} - V_i \right) e^{-\frac{t-t_p}{RC}} \\ &= (V_p - V_i) e^{-\frac{t-t_p}{RC}} \end{aligned}$$

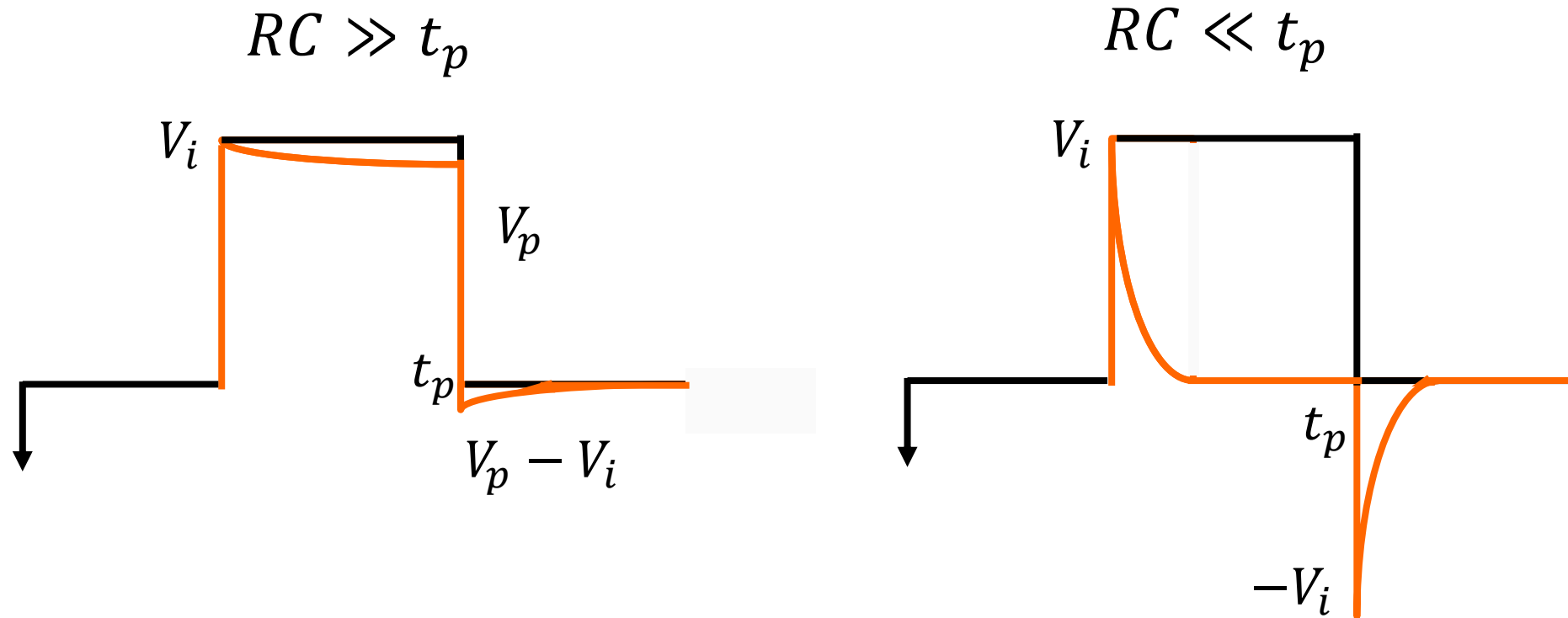
where

$$V_p = V_i e^{-\frac{t_p}{RC}}$$



High Pass RC Circuit

Pulse Input



High Pass RC Circuit

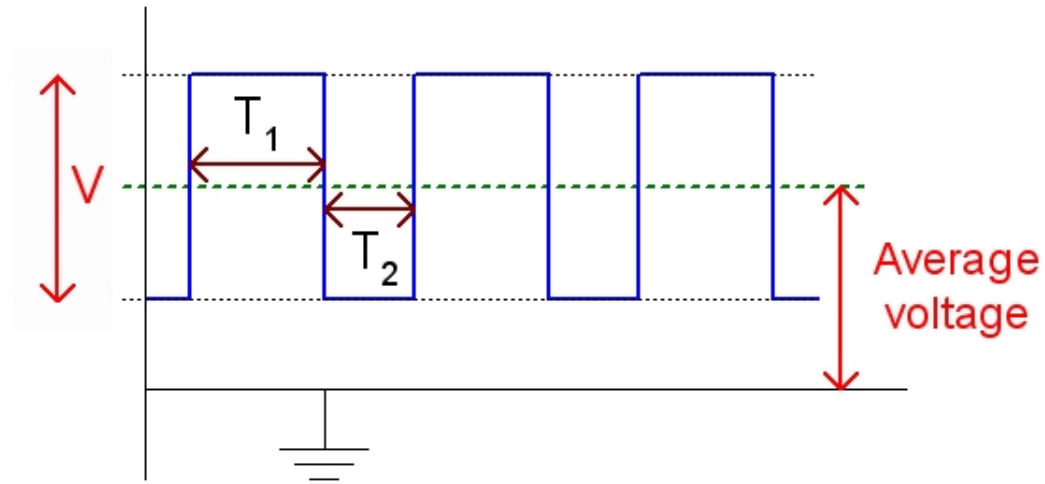
Behavior

- The average level of the output signal is always zero (DC block).
- When the input changes discontinuously by V , the output changes discontinuously by the same amount in the same direction.
- During any finite time interval when the input maintains a constant level, the output decays exponentially towards zero voltage.

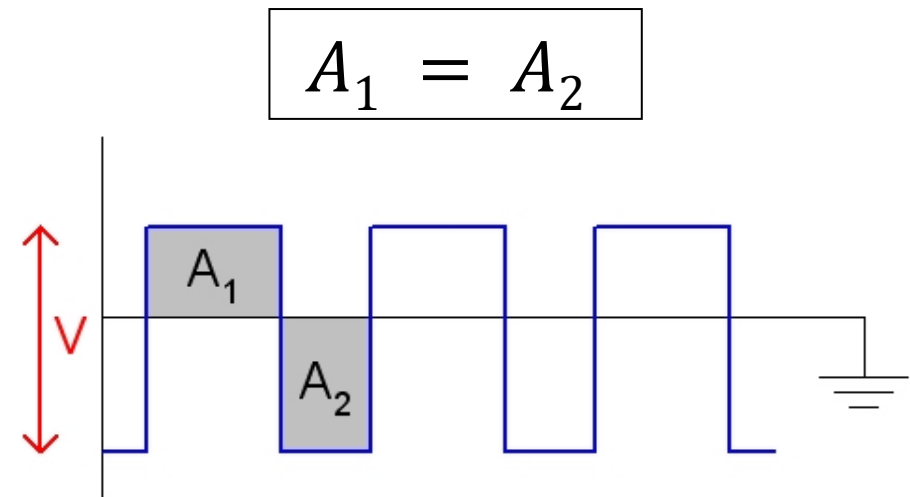
High Pass RC Circuit

Rectangular Wave Input

When $RC \gg T_1, T_2$



Input

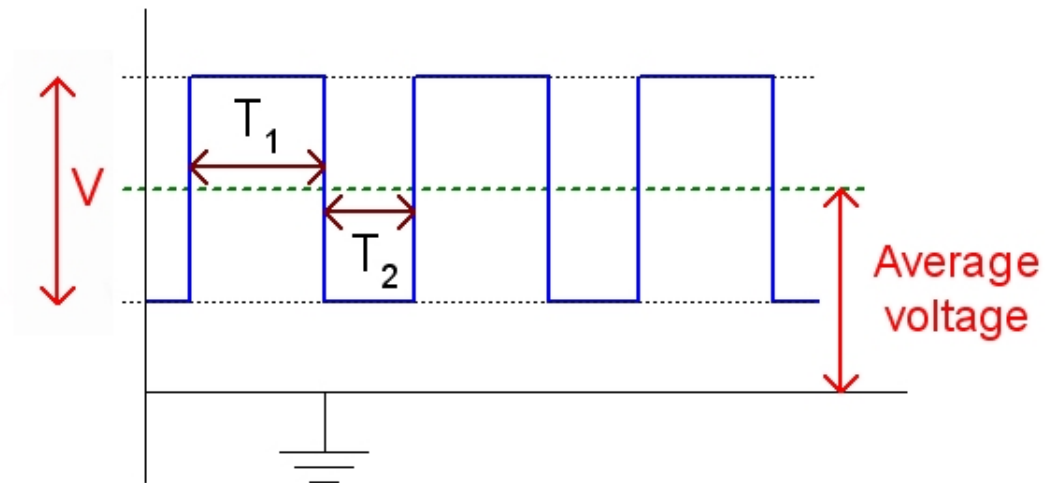


Output

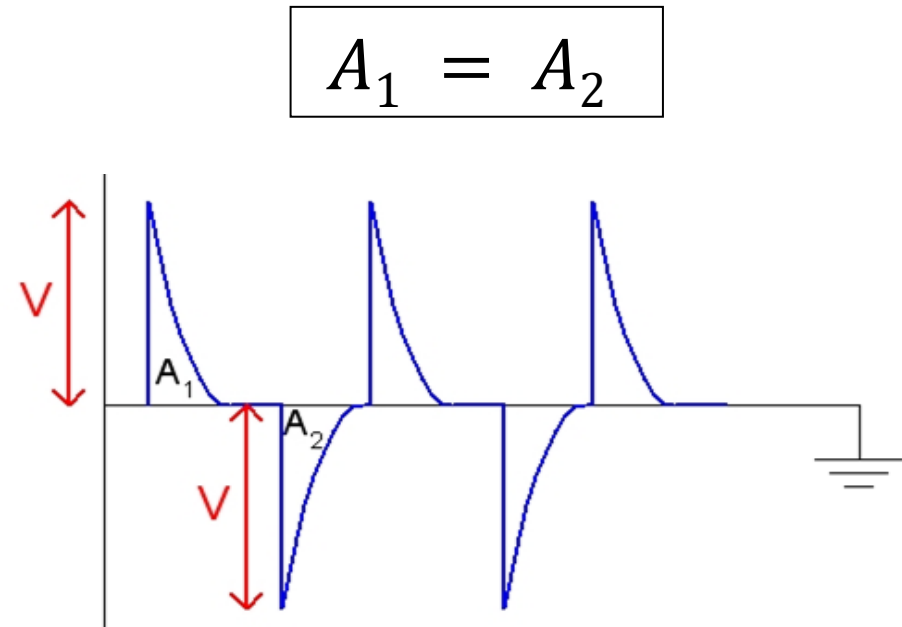
High Pass RC Circuit

Rectangular Wave Input

When $RC \ll T_1, T_2$



Input

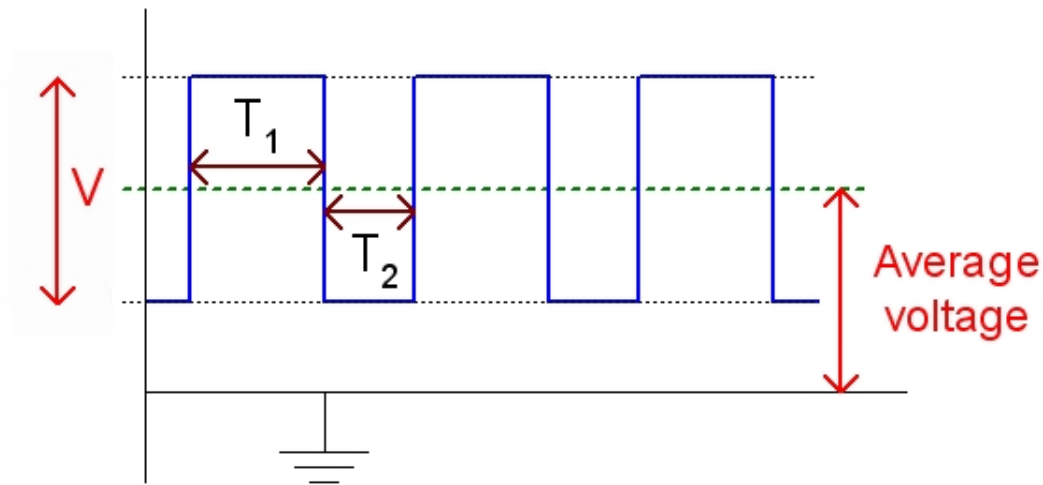


Output

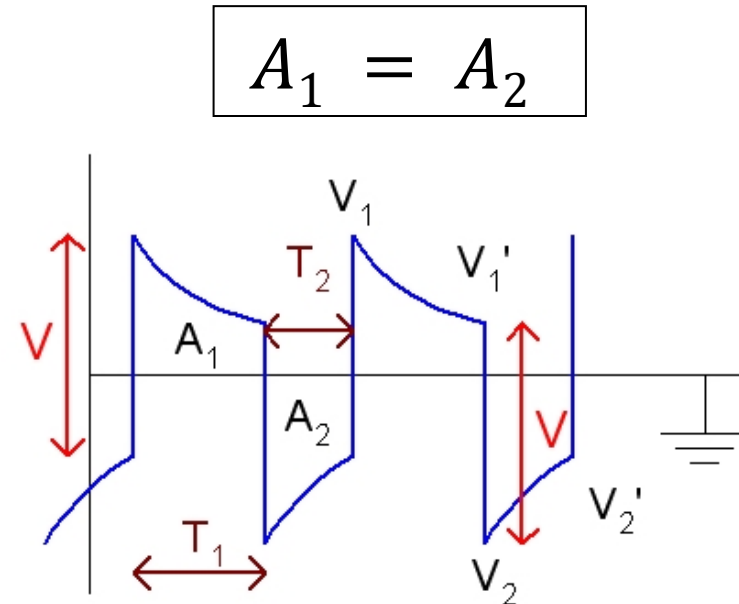
High Pass RC Circuit

Rectangular Wave Input

When RC is comparable to T_1 and T_2



Input

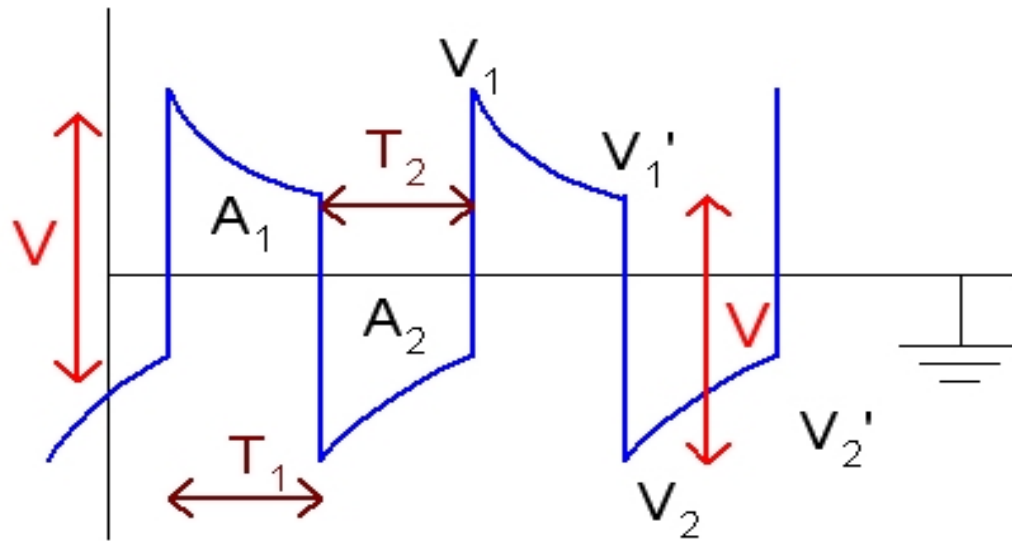


Output

High Pass RC Circuit

Square Wave Input

When RC is comparable to T_1 and T_2



Output

A $V_1' = V_1 e^{-\frac{T_1}{RC}}$

B $V_2' = V_2 e^{-\frac{T_2}{RC}}$

C $V_1' - V_2 = V$

D $V_1 - V_2' = V$

E $V_1 = -V_2$

F $V_1' = -V_2'$

where $T_1 = T_2 = \frac{T}{2}$

High Pass RC Circuit

Square Wave Input

Derive the formulas for V_1, V_1', V_2, V_2' in terms of the known variables V, R , and C .

From **D**, **F**, **A**, and **E**,

$$V_1 - V_2' = V$$

$$V_1 + V_1' = V$$

$$V_1 + V_1 e^{-\frac{T}{2RC}} = V$$

$$\mathbf{G} \rightarrow V_1 = \frac{V}{1 + e^{-\frac{T}{2RC}}} = -V_2$$

From **C**, **G**, and **F**,

$$V_1' - V_2 = V$$

$$V_1' + \frac{V}{1 + e^{-\frac{T}{2RC}}} = V$$

$$V_1' = V - \frac{V}{1 + e^{-\frac{T}{2RC}}}$$

$$V_1' = V \left(\frac{e^{-\frac{T}{2RC}}}{1 + e^{-\frac{T}{2RC}}} \right) = \frac{V}{e^{\frac{T}{2RC}} (1 + e^{-\frac{T}{2RC}})} = \frac{V}{1 + e^{\frac{T}{2RC}}}$$

$$V_1' = \frac{V}{1 + e^{\frac{T}{2RC}}} = -V_2'$$

High Pass RC Circuit

Square Wave Input

Case: $\frac{T}{2} \ll RC$

The Taylor series expansion for an exponential function is

$$e^x = 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \dots$$

$$e^x = 1 + x \quad \text{if } x < 0.1$$

$$\text{If } x = -\frac{T}{2RC} < 0.1,$$

$$e^{-\frac{T}{2RC}} = 1 - \frac{T}{2RC}$$

$$V_1 = \frac{V}{1 + e^{-\frac{T}{2RC}}} = \frac{V}{1 + 1 - \frac{T}{2RC}} = \frac{V}{2 - \frac{T}{2RC}} = \frac{V}{2} \left(\frac{1}{1 - \frac{T}{4RC}} \right)$$

Using binomial approximation $\left(\frac{1}{1-x} = 1 + x \quad \text{for } |x| \ll 1 \right)$,

$$V_1 \approx \frac{V}{2} \left(1 + \frac{T}{4RC} \right) = -V_2$$

Similarly,

$$V_1' = \frac{V}{1 + e^{\frac{T}{2RC}}} = \frac{V}{1 + 1 + \frac{T}{2RC}} = \frac{V}{2} \left(\frac{1}{1 + \frac{T}{4RC}} \right)$$

$$V_1' \approx \frac{V}{2} \left(1 - \frac{T}{4RC} \right) = -V_2'$$

High Pass RC Circuit

Square Wave Input

Cut-off frequency

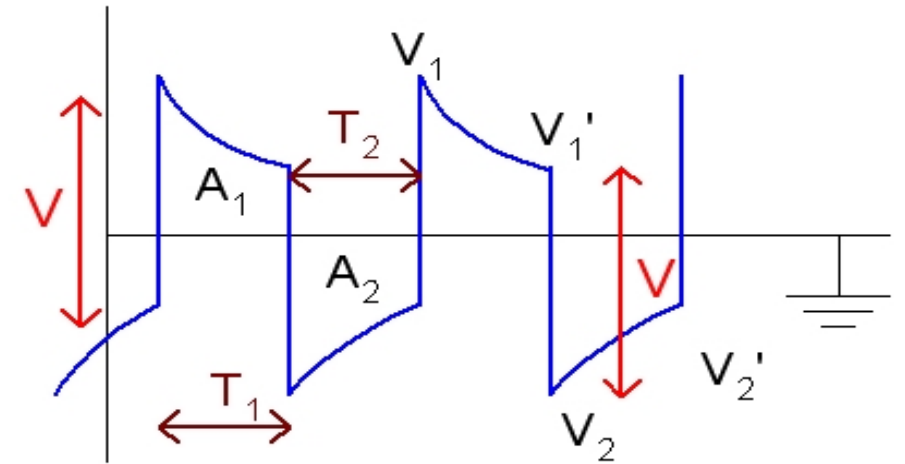
$$f_c = \frac{1}{2\pi RC} \quad \text{or} \quad \omega_c = 2\pi f_c = \frac{1}{RC}$$

Tilt

$$\text{Tilt} = V_1 - V_1' \approx \frac{VT}{4RC}$$

$$\% \text{Tilt} = \frac{\text{Tilt}}{\frac{V}{2}} \times 100\% = \frac{T}{2RC} \times 100\% = \frac{1}{2fRC} \times 100\%$$

$$\% \text{Tilt} = \frac{\pi f_c}{f} \times 100\%$$



High Pass RC Circuit

Square Wave Input

Exercise:

An amplifier has input coupling capacitor of $30\ \mu F$ and input resistance of $20\ k\Omega$. Determine the lowest square wave frequency that can be applied as an input if the output tilt is not to exceed 5%?

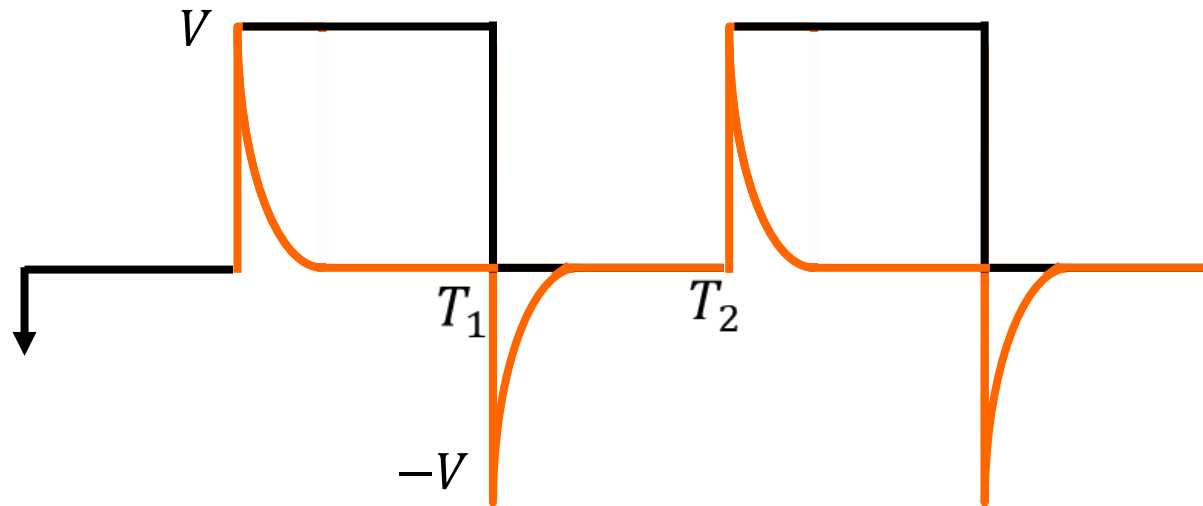
ANS: 16.67 Hz

High Pass RC Circuit

as a Differentiator

If RC is very small compared to the rate at which the input V_i changes, the high pass RC acts as a differentiator.

$$RC \ll T_1, T_2$$



High Pass RC Circuit as a Differentiator

Input: $V_i = V_m \sin \omega t$

Desired Output: $\frac{dV_i}{dt} = \omega V_m \cos \omega t$

Note: i_c leads V_i by θ thus, V_o leads V_i by θ where

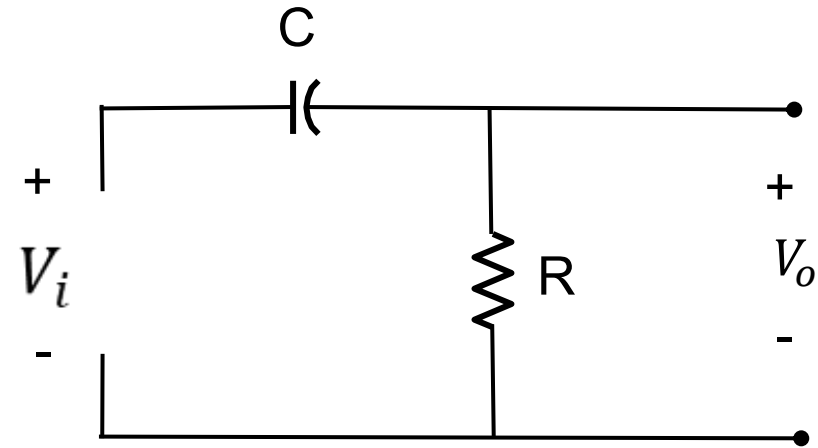
$$\theta = \tan^{-1} \frac{X_c}{R} = \tan^{-1} \frac{1}{\omega RC}$$

By voltage division,

$$V_o = \frac{R}{\sqrt{R^2 + \left(\frac{1}{\omega C}\right)^2}} V_m \sin(\omega t + \theta)$$

If ωRC is very small, then $\theta \approx 90^\circ$

$$V_o \approx \omega RC V_m \cos \omega t$$

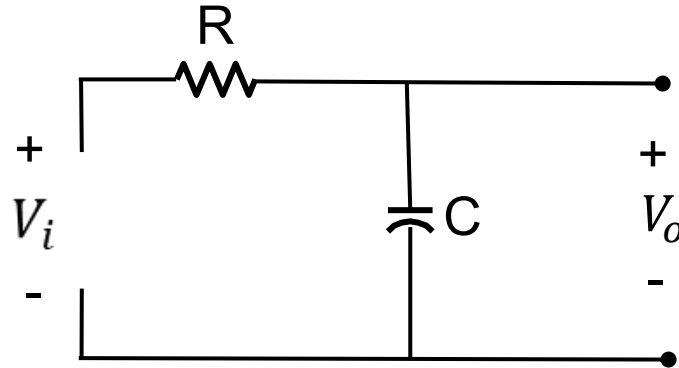


Low Pass RC Circuit

Sinusoidal Input

$$V_i = A \cos(2\pi f t + \varphi)$$

$$X_c = \frac{1}{\omega C}$$



Parameters

- upper 3dB frequency is $f_H = \frac{1}{2\pi RC}$

- gain is $|H(f)| = \frac{1}{\sqrt{1 + \left(\frac{f}{f_H}\right)^2}}$



- the output leads the input by an angle of $\theta = \tan^{-1} \left(\frac{f}{f_H} \right)$

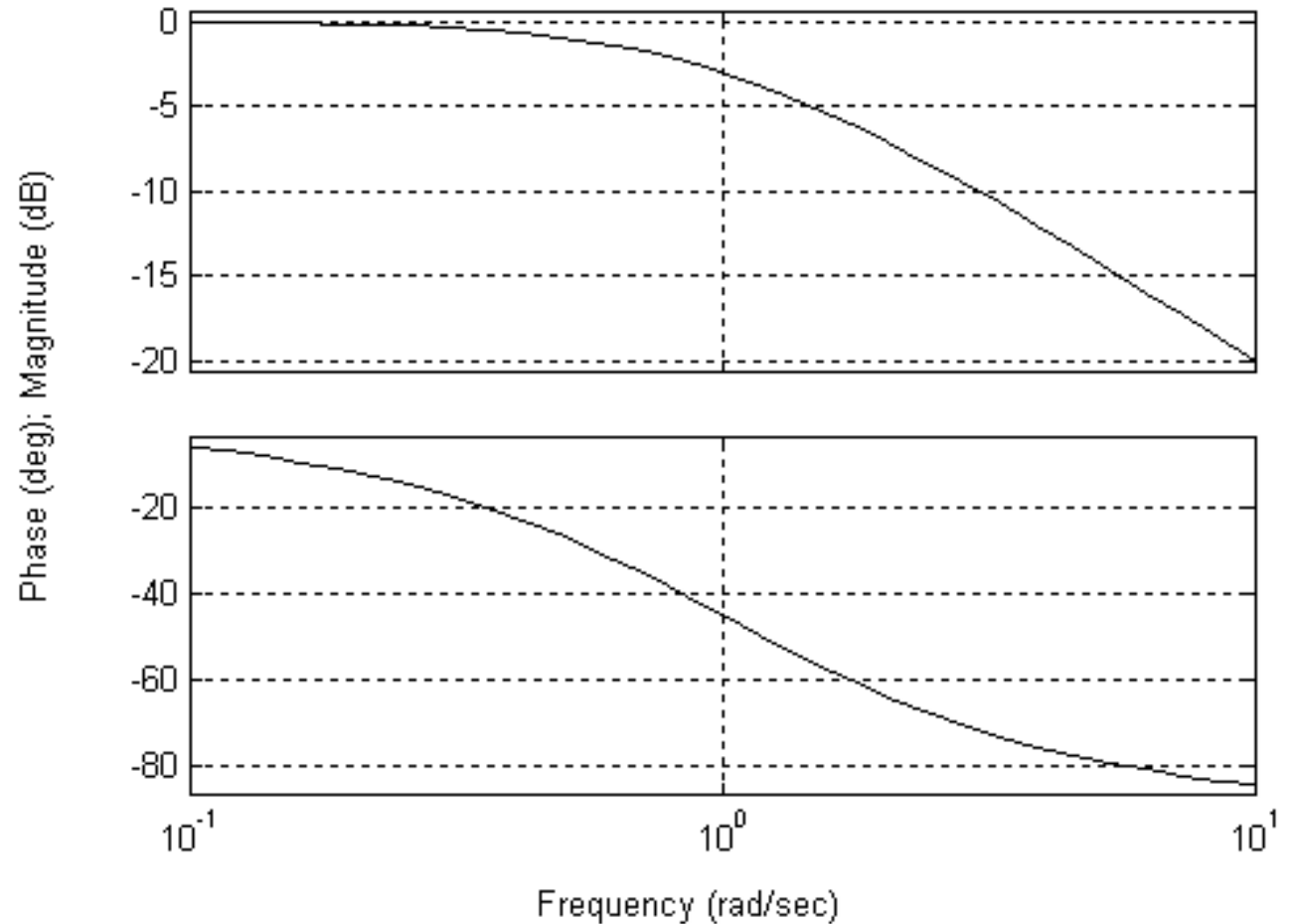
Low Pass RC Circuit

Sinusoidal Input

Frequency Response

At $f = f_c$,

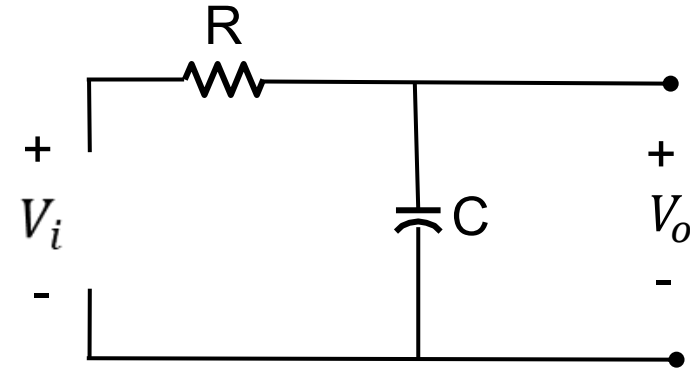
- $|H(f_c)| = 0.707$
- $X_c = R$



Low Pass RC Circuit

Step Voltage Input

$$V_o = V_o(\infty) + [V_o(0^+) - V_o(\infty)]e^{-\frac{t}{RC}}$$

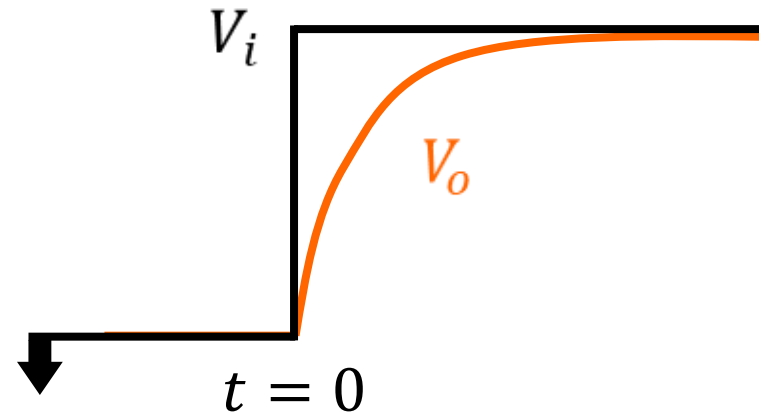


Steady-state analysis will show that

- $V_o(0^+) = 0$
- $V_o(\infty) = V_i$

Hence,

$$V_o = V_i \left(1 - e^{-\frac{t}{RC}}\right)$$



rise time: $t_R = \frac{0.35}{f_H} = 2.2RC$

Low Pass RC Circuit

Pulse Input

At $0 < t < t_p$

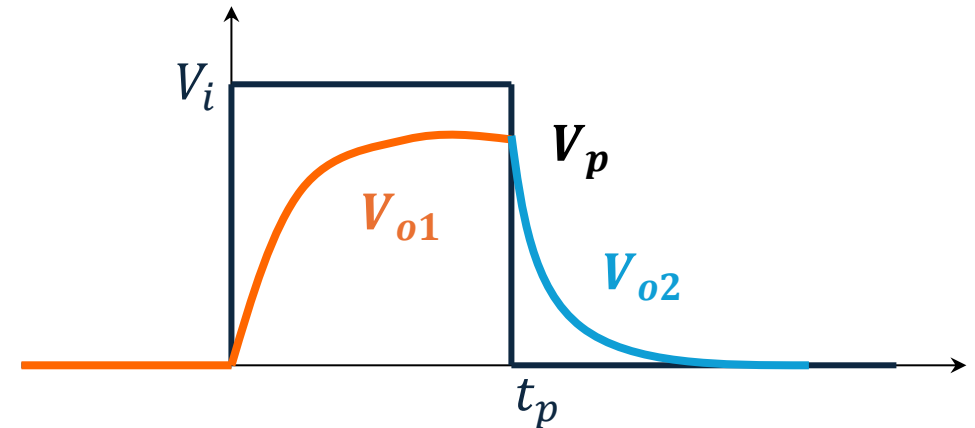
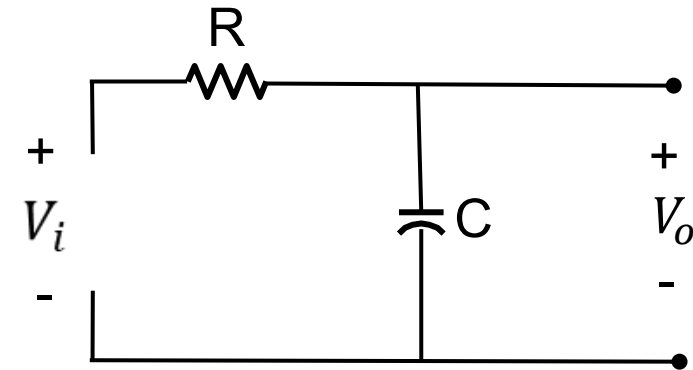
$$V_{o1} = V_i \left(1 - e^{-\frac{t}{RC}}\right)$$

At $t > t_p$

$$V_{o2} = V_p e^{-\frac{t-t_p}{RC}}$$

where

$$V_p = V_i \left(1 - e^{-\frac{t_p}{RC}}\right)$$



Note: As a rule of thumb, f_H should be $> \frac{1}{t_p}$ to minimize distortion (preserve the shape of the pulse).

Low Pass RC Circuit

Rectangular Wave Input

$$V_{o1} = V' + (V_1 - V')e^{-t/RC}$$

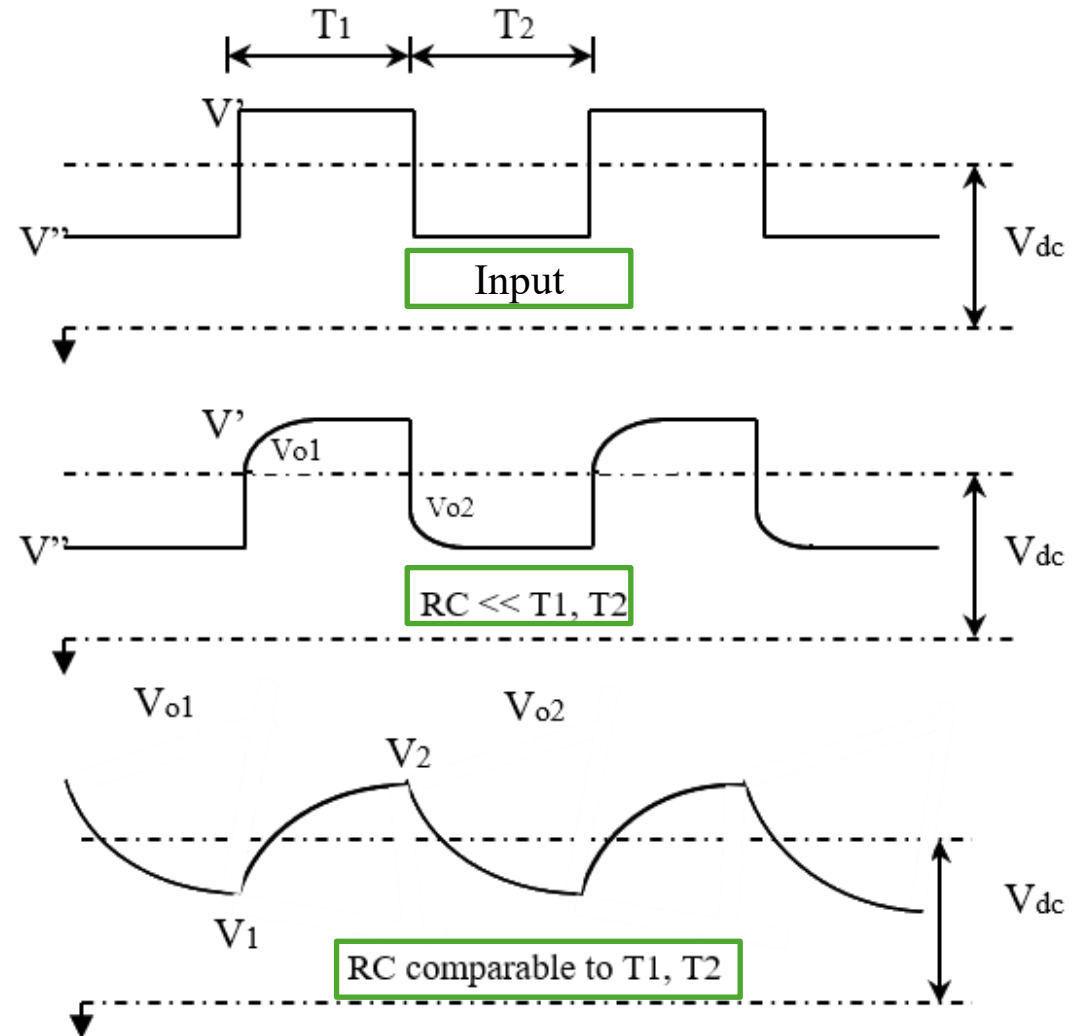
$$V_{o2} = V'' + (V_2 - V'')e^{-(t-T_1)/RC}$$

at $t = T_1$, $V_{o1} = V_2$ and

at $t = T_2$, $V_{o2} = V_1$

$$\therefore V_2 = V' + (V_1 - V')e^{-T_1/RC}$$

$$V_1 = V'' + (V_2 - V'')e^{-T_2/RC}$$

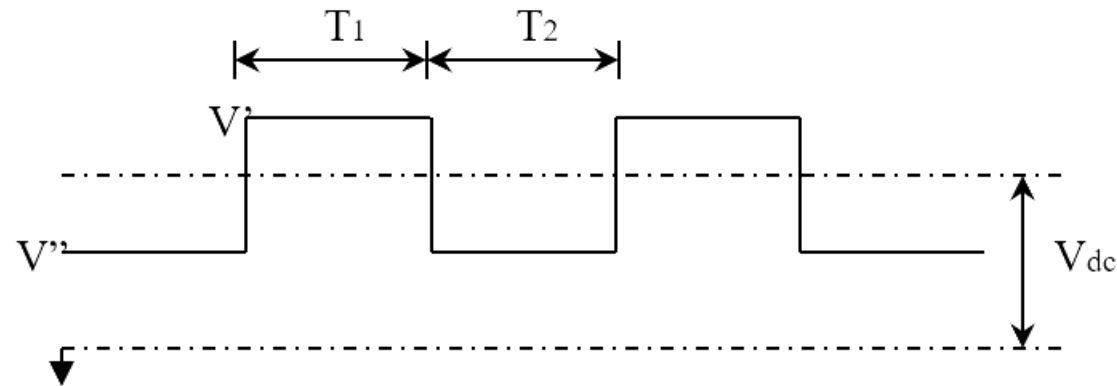


Low Pass RC Circuit

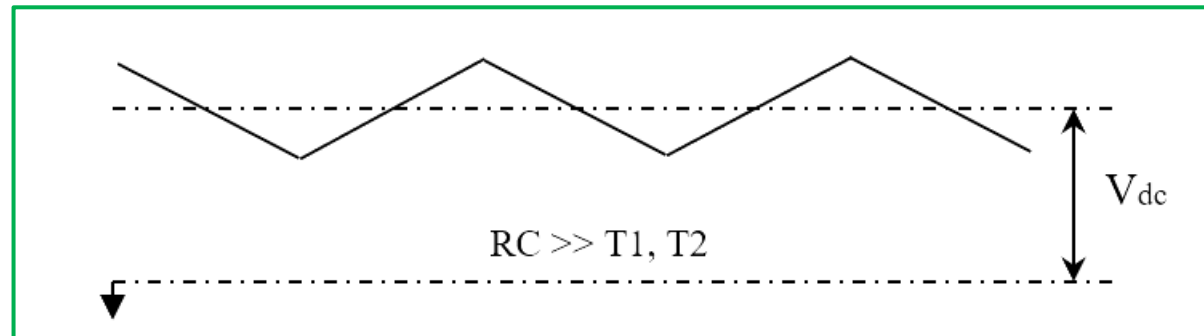
Rectangular Wave Input

Exercise: Draw the waveform of the output voltage if $RC \gg T_1, T_2$.

Input:



Output:



Low Pass RC Circuit

as an Integrator

When RC is large compared to the time the input V_i can make an appreciable change, the circuit becomes an integrator.

$$RC \gg T_1, T_2$$

→ voltage drop across C is negligible compared to the drop across R .

$$\begin{aligned} V_o &= \frac{1}{C} \int i_c \, dt \cong \frac{1}{C} \int \frac{V_R}{R} \, dt \\ &\cong \frac{1}{RC} \int V_i \, dt \end{aligned}$$

Summary

- HP RC acts as high-pass filter with lower 3-dB cutoff frequency of

$$f_L = f_c = \frac{1}{2\pi RC}$$

- LP RC acts as a low-pass filter with upper 3-dB cutoff frequency of

$$f_H = f_c = \frac{1}{2\pi RC}$$

- Linear waveshaping describes how passive RC (and RL) networks alter non-sinusoidal waveforms based on the circuit's time constant (τ):
 - HP RC differentiates inputs when $RC \ll T$
 - LP RC integrates inputs when $RC \gg T$.